

SOCIOL 723: Social Statistics II

Week 3, Maximum Likelihood: Theory

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Today

Why We Need More Than Least Squares

The Likelihood Function

Finding the Maximum

Information and the Variance of the MLE

Properties of the MLE

Likelihood-Based Tests

★ = essential, you must understand this

★ = for understanding, not examined

Why Likelihood

What Least Squares Bought Us, and What It Cost ★

Two weeks, one idea: pick the line that makes the squared misses as small as possible. Look at how much that bought:

- A slope with a clear meaning (the best linear approximation to the CEF), computable with sums alone.
- Honest uncertainty for it, robust to heteroskedasticity and clustering, without ever assuming what the errors look like.

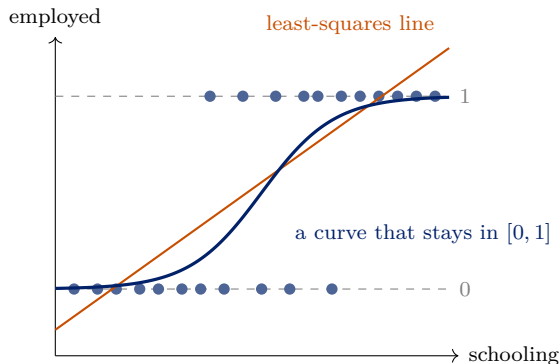
It asked almost nothing in return: never *how* Y was generated, only *which line fits best*.

That is also its limit

Least squares cannot ask **how were these outcomes produced?** For earnings you can afford not to ask. For much of sociology's data, you cannot.

Where the Line Runs Out ★

Regress “employed” (0/1) on years of schooling, by least squares:



- The line predicts “probabilities” below 0 and above 1: the shape of the outcome carries information the line cannot use.
- Counts go negative the same way; durations get censored; categories have no order. The problem is not the slope: *a line is the wrong kind of answer.*

The Question Underneath ★

Least squares asks which line comes closest to the points. Maximum likelihood asks something different:

“Which parameter values would have made the data I observed most likely?”

To ask that, you have to say how Y is produced: a coin for employment, a bell curve centred on a line for earnings. In return:

- One method for any outcome: binary, count, ordered, duration.
- Standard errors from the shape of the likelihood, not a new formula per model.
- One way to test hypotheses, logit or Poisson alike: three tests read off the same curve.
- Nothing lost: with bell-shaped errors it returns the least-squares line exactly. (Call this *the hook*; we cash it at the end of the next section.)

Hence two weeks: this week the method, next week logit, probit, Poisson.

Two Stances, Both Legitimate

Two questions, two stances toward a regression; this course uses both.

- **Design-based** (Weeks 1–2, 5–11): β is the best linear approximation to the CEF; assume little about how Y was generated; let the *research design* carry the causal claim.
- **Model-based** (Weeks 3–4): write a probability model for Y and estimate the parameter values that make the data most probable. Needed when the outcome is not a continuous, unbounded number.

Neither is “the right one”

Say which stance a result rests on: a logit coefficient and a difference-in-differences estimate answer different kinds of question.

The Likelihood

Probability Versus Likelihood ★

Both start from the same object, the density (or mass function) $f(y | \theta)$. What differs is *what you hold fixed*.

Probability

$$f(y | \theta)$$

θ is *fixed and known*; y varies.

“If the coin is fair, how likely am I to see 7 heads in 10 tosses?”

Likelihood

$$\mathcal{L}(\theta; y) \equiv f(y | \theta)$$

y is *fixed* (it is your data); θ varies.

“Given that I saw 7 heads in 10 tosses, which values of θ make that outcome most probable?”

The single most important sentence today

The likelihood is *the same formula read backwards*: a function of θ with the data fixed. It is **not** a distribution over θ : it does not integrate to one.

A First Example: Marbles in a Bag ★

An opaque bag holds blue and white marbles; θ is the proportion blue. You draw 10 marbles *with replacement* and get 7 blue. Ten draws are ten *observations*, each a Bernoulli outcome (blue or not); the summary “7 of 10” is a binomial count. Build its probability step by step:

- **One draw:** blue with probability θ , white with probability $1 - \theta$ (replacement makes the draws identical and independent).
- **One particular sequence**, say B B B B B B W W W: independence means multiply, $\theta \cdot \theta \cdots \theta \cdot (1 - \theta)(1 - \theta)(1 - \theta) = \theta^7(1 - \theta)^3$. Any other order with 7 blues and 3 whites has the *same* product.
- **All such sequences:** there are $\binom{10}{3} = \binom{10}{7} = \frac{10!}{3!7!} = 120$ ways to place the 3 whites (equivalently the 7 blues) among the 10 draws, so $\Pr(7 \text{ blue} \mid \theta) = 120 \theta^7(1 - \theta)^3$. ($\binom{n}{k}$, read “ n choose k ,” is the standard notation.)

The 120 does not involve θ : it cannot change which candidate wins.

Marbles: Reading It as a Likelihood

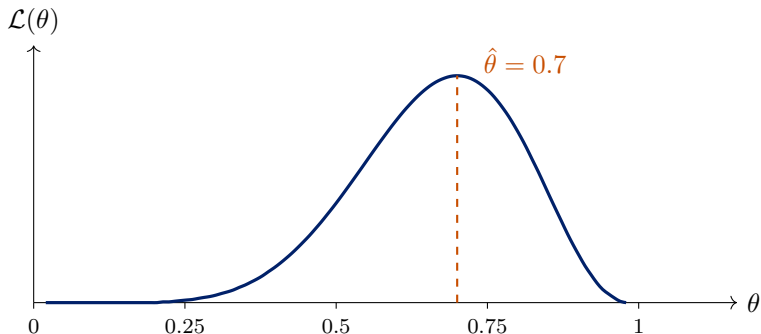
Now hold the data fixed, 7 blue in 10, and let θ vary:

$$\mathcal{L}(\theta) = 120 \theta^7 (1 - \theta)^3.$$

- Try $\theta = 0.3$: $\mathcal{L} = 120(0.3)^7(0.7)^3 \approx 0.0090$.
- Try $\theta = 0.5$: $\mathcal{L} = 120(0.5)^{10} \approx 0.117$.
- Try $\theta = 0.7$: $\mathcal{L} = 120(0.7)^7(0.3)^3 \approx 0.267$.

Among these three, $\theta = 0.7$ makes the data we actually saw most probable. The MLE is the value that does best among *all* $\theta \in [0, 1]$: the next frame plots the whole curve.

The Likelihood, Plotted



- The height at any θ is the probability of *these* data if that θ were true.
- The peak is the MLE. The **width** of the peak will turn out to be our measure of uncertainty; that is the whole of Section 4.

From One Observation to Many ★

With n observations we need the *joint* density $f(y_1, \dots, y_n \mid \theta)$. If the observations are independent given θ , it factors:

$$\mathcal{L}(\theta; y_1, \dots, y_n) = \prod_{i=1}^n f(y_i \mid \theta).$$

(Two observations first: independence means $f(y_1, y_2 \mid \theta) = f(y_1 \mid \theta) f(y_2 \mid \theta)$, the probability of both is the product; n observations just repeat the step.)

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(Two observations first: independence means $f(y_1, y_2 \mid \theta) = f(y_1 \mid \theta) f(y_2 \mid \theta)$, the probability of both is the product; n observations just repeat the step.) A product of hundreds of numbers below one rounds to zero on a computer and is awkward to differentiate, so we take logs:

$$\ell(\theta) \equiv \log \mathcal{L}(\theta) = \sum_{i=1}^n \log f(y_i \mid \theta)$$

- log is strictly increasing, so it **does not move the maximum**.
- A product becomes a sum, so the derivative is a sum over observations: setting it to zero is a kind of averaging.

Worked Example 1: Bernoulli Draws ★

The marble game: n Bernoulli draws, $Y_i \in \{0, 1\}$, $\Pr(Y_i = 1) = \theta$. We observe the n draws and estimate θ , the probability of a success. One draw's mass function as a single formula:

$$f(y_i | \theta) = \theta^{y_i} (1 - \theta)^{1-y_i}.$$

(At $y_i = 1$ it gives θ ; at $y_i = 0$, $1 - \theta$.)

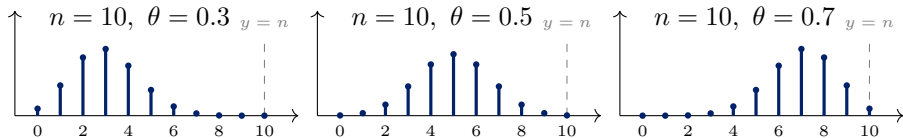
$$\mathcal{L}(\theta) = \prod_{i=1}^n \theta^{y_i} (1 - \theta)^{1-y_i} = \theta^{\sum_i y_i} (1 - \theta)^{n - \sum_i y_i},$$

$$\ell(\theta) = \left(\sum_i y_i \right) \log \theta + \left(n - \sum_i y_i \right) \log(1 - \theta).$$

Sequence or count? This is the probability of the *particular* 0/1 sequence; the binomial multiplies it by $\binom{n}{\sum_i y_i}$ for the *count*. A constant factor, so the same MLE: the total $\sum_i y_i$ is all the data can tell you about θ .

Worked Example 1: Reading the Binomial ★

$$\Pr(Y = y \mid n, \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}, \quad y = 0, 1, \dots, n$$



- **The pieces:** θ^y for the y successes, $(1 - \theta)^{n-y}$ for the failures, $\binom{n}{y}$ for the orders they can come in.
- **The shape:** mean $n\theta$, variance $n\theta(1 - \theta)$ (n independent draws with mean θ and variance $\theta(1 - \theta)$; both add), and a hard ceiling at n .
- **Fix the data, compare the panels:** fix $y = 7$; which panel puts the most mass on 7? The $\theta = 0.7$ panel. Data fixed, θ varying: the likelihood.

Counting Units or Counting Events? ★

Sociology's counts come in two kinds.

- **Units with an event, out of a known n :** 12 of 30 students protested. Each unit is a Bernoulli 0/1 (Worked Example 1); the total is binomial; it cannot exceed n .
- **Events one unit accumulates over exposure:** arrests of one person in a year; children; publications. There is no n to count out of, and no ceiling.

Rule of thumb

First kind: binomial (in practice a logit on the 0/1, Week 4). Second kind: we need a distribution with no ceiling. Build it from the binomial, next.

Worked Example 2: From the Binomial to the Poisson ★

Start from the binomial count of Worked Example 1:

$$\Pr(Y = y \mid n, p) = \binom{n}{y} p^y (1 - p)^{n-y}.$$

Read it as *arrests of one person in a year*: slice the year into n moments, each a Bernoulli trial with chance p of an arrest. Then y is the number of arrests: p^y for the y moments with one, $(1 - p)^{n-y}$ for the rest, $\binom{n}{y}$ for which moments they were.

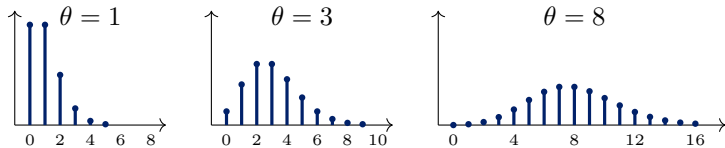
Now slice finer: 365 days, 8,760 hours, 525,600 minutes, ... The slices $n \rightarrow \infty$ and the chance per slice $p \rightarrow 0$, but the expected arrests per year, $np = E[Y] = \theta$, cannot depend on the slicing. Fix y and take the limit:

$$\binom{n}{y} p^y (1 - p)^{n-y} \longrightarrow \frac{\theta^y e^{-\theta}}{y!}, \quad y = 0, 1, 2, \dots$$

The **Poisson distribution**: one parameter θ (the letter is recycled; it always means “the parameter”), the expected count. Neither n nor p survives, so nothing caps the count. Why ★: $p = \theta/n$, so $\binom{n}{y} p^y \rightarrow \theta^y / y!$ and $(1 - \theta/n)^{n-y} \rightarrow e^{-\theta}$.

Worked Example 2: Reading the Poisson ★

$$\Pr(Y = y \mid \theta) = \frac{\theta^y e^{-\theta}}{y!}, \quad y = 0, 1, 2, \dots$$



- **The pieces:** θ^y grows with y , $y!$ shrinks it back (large counts are rare), $e^{-\theta}$ makes the probabilities sum to one.
- **The shapes:** piled at zero and right-skewed for small θ ; nearly a bell centred at θ for large θ . No ceiling anywhere: the binomial's n is gone.
- **The signature:** mean and variance are both θ (inherited from the binomial: $np \rightarrow \theta$ and $np(1-p) \rightarrow \theta$ as $p \rightarrow 0$). Spread grows with the mean.

Worked Example 2: The Poisson Log-Likelihood ★

Independent Poisson counts $Y_1, \dots, Y_n$, common θ ; step by step:

$$\mathcal{L}(\theta) = \prod_{i=1}^n \frac{\theta^{y_i} e^{-\theta}}{y_i!}$$

independence: multiply

$$\ell(\theta) = \sum_{i=1}^n \log \left(\frac{\theta^{y_i} e^{-\theta}}{y_i!} \right)$$

log turns the product into a sum

$$= \sum_{i=1}^n \left(y_i \log \theta - \theta - \log(y_i!) \right)$$

$\log \theta^y = y \log \theta$, $\log e^{-\theta} = -\theta$

$$= \log(\theta) \sum_i y_i - n\theta - \sum_i \log(y_i!)$$

collect: n copies of θ

- The last term has no θ : an *additive constant*, droppable for maximization but not when comparing log-likelihood *values* across models.
- Again the data enter only through $\sum_i y_i$: the total count is sufficient.

Worked Example 3: The Normal Linear Model ★

The normal (Gaussian) density, a bell centred at μ with spread σ :

$$f(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right).$$

One assumption from Week 1 and one new one, with σ^2 held fixed:

1. **Linear CEF:** $E[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$.
2. **Normal errors:** $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, independently.

Together they say $Y_i \mid X_i \sim \mathcal{N}(\beta_0 + \beta_1 X_i, \sigma^2)$: the bell's centre μ is the line. Put $\mu = b_0 + b_1 X_i$ into the density, and one observation's likelihood at candidate values (b_0, b_1) is

$$f(Y_i \mid b_0, b_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - b_0 - b_1 X_i)^2}{2\sigma^2}\right).$$

Cashing the Hook: OLS Is an MLE ★

Product over independent observations, then log:

$$\begin{aligned}\ell(b_0, b_1) &= \sum_{i=1}^n \log f(Y_i \mid b_0, b_1) = \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(Y_i - b_0 - b_1 X_i)^2}{2\sigma^2} \right] \\ &= \underbrace{-\frac{n}{2} \log(2\pi\sigma^2)}_{\text{no } b_0, b_1} - \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2}_{\text{Week 1's } S(b_0, b_1)}.\end{aligned}$$

- The candidates enter only through S , with a minus sign and a positive factor in front. **Maximizing ℓ is exactly minimizing S :** the MLE is the least-squares line, digit for digit.
- Least squares was never arbitrary: it is what “make the data as probable as possible” prescribes for normal errors around a line.

What the Hook Does Not Say ★

- **OLS itself never needed normality.** Weeks 1–2 built it from moments alone; its large-sample inference came from the CLT. Normality entered only to make the t table exact.
- **Normality is the price of admission** to the likelihood framework. For a continuous outcome it buys little: OLS already gave the estimate, and Week 2 already gave its standard errors.

The point of the exercise

Not “assume normality,” but “one principle contains least squares as a special case and extends to everything least squares cannot do.”

Preview: Where the Regressors Go ★

So far every observation shares one θ . Next week each observation gets its own, driven by covariates:

$$\theta_i = \exp(\beta_0 + \beta_1 X_i), \quad \ell(\beta_0, \beta_1) = \sum_i \left[y_i(\beta_0 + \beta_1 X_i) - e^{\beta_0 + \beta_1 X_i} - \log(y_i!) \right].$$

- The exponential keeps every θ_i positive, whatever the covariates do; this is the *log link* of Poisson regression.
- Same recipe: density, product, log, sum. Only the parameter changed from a number to a function of X_i .
- No closed form any more (β sits inside the exponential), so the maximum is found numerically: Section 3 does it by hand.

Finding the MLE

The Score Function ★

Definition

The **score** is the derivative of the log-likelihood with respect to the parameter:

$$S(\theta) \equiv \frac{\partial \ell(\theta)}{\partial \theta} = \sum_{i=1}^n \frac{\partial \log f(y_i | \theta)}{\partial \theta}.$$

- The MLE $\hat{\theta}$ solves $S(\hat{\theta}) = 0$, the **first-order condition**, exactly as the normal equations were the FOC for least squares.
- Second-order condition: $\partial^2 \ell / \partial \theta^2 < 0$ at $\hat{\theta}$, so that we have a maximum rather than a minimum.
- Write θ_0 for the *true* value. A population fact, derived in Section 4: across repeated samples, $E[S(\theta_0)] = 0$ (the true θ_0 fixed, the data varying: Week 2's sampling-distribution view).

Bernoulli: Deriving $\hat{\theta}$ ★

$$\ell(\theta) = \left(\sum_i y_i\right) \log \theta + \left(n - \sum_i y_i\right) \log(1 - \theta)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1 - \theta} \stackrel{!}{=} 0.$$

Bernoulli: Deriving $\hat{\theta}$ ★

$$\ell(\theta) = \left(\sum_i y_i \right) \log \theta + \left(n - \sum_i y_i \right) \log(1 - \theta)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1 - \theta} \stackrel{!}{=} 0.$$

Multiply through by $\theta(1 - \theta)$:

$$(1 - \theta) \sum_i y_i - \theta \left(n - \sum_i y_i \right) = 0$$

$$\sum_i y_i - \theta \sum_i y_i - n\theta + \theta \sum_i y_i = 0$$

$$\boxed{\hat{\theta} = \frac{1}{n} \sum_i y_i = \bar{y}}$$

The MLE of a probability is the sample proportion, derived, not assumed.

Poisson: Deriving $\hat{\theta}$ ★

Differentiate the log-likelihood and set the score to zero:

$$\ell(\theta) = \log(\theta) \sum_i y_i - n\theta - \sum_i \log(y_i!)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - n$$

$\frac{d}{d\theta} \log \theta = \frac{1}{\theta}$; the constant's derivative is 0

$$S(\hat{\theta}) = 0 \implies \frac{\sum_i y_i}{\hat{\theta}} = n \implies \boxed{\hat{\theta} = \frac{\sum_i y_i}{n} = \bar{y}}$$

Second-order condition, to be sure it is a maximum:

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} < 0. \quad \checkmark$$

Both examples give $\bar{y}$, which is not a coincidence: for these distributions the mean *is* the parameter, and maximum likelihood finds it.

Normal Linear Model: Deriving $\hat{\beta}$ and $\hat{\sigma}^2$ ★

Three parameters, so three first-order conditions. With $e_i = Y_i - b_0 - b_1 X_i$:

$$\frac{\partial \ell}{\partial b_0} = \frac{1}{\sigma^2} \sum_i e_i \stackrel{!}{=} 0, \quad \frac{\partial \ell}{\partial b_1} = \frac{1}{\sigma^2} \sum_i X_i e_i \stackrel{!}{=} 0 \quad \text{Week 1's normal equations, exactly}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i e_i^2 \stackrel{!}{=} 0 \quad \implies \quad \hat{\sigma}^2 = \frac{1}{n} \sum_i \hat{e}_i^2 \quad \text{solve; } \hat{e}_i \text{ are the OLS residuals}$$

The first two conditions are Week 1's normal equations, so $\hat{\beta}_0, \hat{\beta}_1$ are the OLS estimates: the hook, cashed a second way.

The MLE of σ^2 is biased

It divides by n , not $n - 2$: $E[\hat{\sigma}^2] = \frac{n-2}{n}\sigma^2 < \sigma^2$. Week 1's s^2 was the bias-corrected version.

When There Is No Closed Form

Most likelihoods you will meet, logit, probit, negative binomial, anything with covariates, have no algebraic solution to $S(\theta) = 0$. We search numerically.

Newton–Raphson

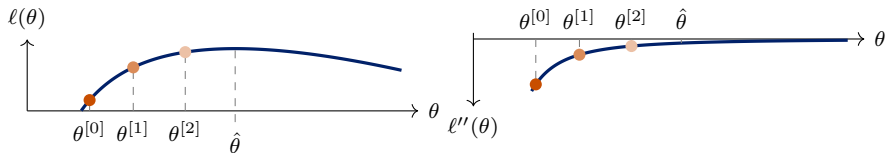
Start at a guess $\theta^{[0]}$ (superscript in brackets: an iteration count, not the true value) and iterate

$$\theta^{[t+1]} = \theta^{[t]} - \left[\frac{\partial^2 \ell}{\partial \theta^2} \bigg|_{\theta^{[t]}} \right]^{-1} S(\theta^{[t]}),$$

a step of size score over curvature. The next frame draws it.

Variants: **BFGS** approximates the second derivative; **Fisher scoring** uses its expectation (what `glm()` does).

Finding the Peak: Newton–Raphson in a Picture



guess θ	1.50	2.55	3.80	4.71
slope S	2.33	0.96	0.32	0.06
$-\ell''$	2.22	0.77	0.35	0.23
step $S/(-\ell'')$	1.05	1.25	0.91	0.27

- Poisson, one observation $y = 5$: $\ell = 5 \log \theta - \theta$, right panel $\ell'' = -5/\theta^2$. Step = slope $\div (-\ell'')$: gradient ascent with the step set by the curvature.
- The step is a *ratio* of two things that both change with θ ; it stops when the slope is zero. $-\ell''(\hat{\theta})$ is the observed information: next section.

Poisson by Hand: Newton–Raphson ★

Five counts: 2, 0, 3, 1, 4. So $n = 5$, $\sum y_i = 10$, and we know the answer, $\hat{\theta} = \bar{y} = 2$. Pretend we do not.

$$\ell(\theta) = 10 \log \theta - 5\theta + \text{const}, \quad S(\theta) = \frac{10}{\theta} - 5, \quad \ell''(\theta) = -\frac{10}{\theta^2}.$$

Start at a guess $\theta^{[0]} = 1$ and step by slope over curvature, $\theta^{[t+1]} = \theta^{[t]} - S/\ell''$:

guess θ	slope S	$-\ell''$	step $S/(-\ell'')$	next guess
1.000	5.000	10.00	0.500	1.500
1.500	1.667	4.44	0.375	1.875
1.875	0.333	2.84	0.117	1.992
1.992	0.020	2.52	0.008	2.000

Four steps to 2.000. At the answer $S = 0$ and $-\ell''(2) = 2.5$: the observed information, so $se = 1/\sqrt{2.5} = 0.63$. Check: $\sqrt{\bar{y}/n} = \sqrt{2/5} = 0.63$.

Regressors Inside: β_1 by Hand, the Function ★

Give each count its own mean, $\theta_i = e^{\beta_1 X_i}$ (intercept known to be 0 for now). Four made-up observations, $X = 0, 1, 2, 3$ and $y = 1, 2, 4, 8$. The Poisson log-likelihood is $\sum_i [y_i \log \theta_i - \theta_i]$ plus a constant, and $\log \theta_i = \beta_1 X_i$. Write out all four terms:

$$\ell(\beta_1) = [1 \cdot \beta_1 \cdot 0 - e^0] + [2 \cdot \beta_1 \cdot 1 - e^{\beta_1}] \quad i = 1, 2: X_i = 0, 1; y_i = 1, 2$$

$$+ [4 \cdot \beta_1 \cdot 2 - e^{2\beta_1}] + [8 \cdot \beta_1 \cdot 3 - e^{3\beta_1}] \quad i = 3, 4: X_i = 2, 3; y_i = 4, 8$$

$$= 34\beta_1 - 1 - e^{\beta_1} - e^{2\beta_1} - e^{3\beta_1} \quad 0 + 2 + 8 + 24 = 34$$

$$S(\beta_1) = 34 - e^{\beta_1} - 2e^{2\beta_1} - 3e^{3\beta_1} \quad \text{differentiate: } \frac{d}{d\beta_1} e^{kX\beta_1} = k e^{k\beta_1}$$

$$\ell''(\beta_1) = -e^{\beta_1} - 4e^{2\beta_1} - 9e^{3\beta_1} \quad \text{differentiate again; always negative}$$

$S(\beta_1) = 0$ has no closed form: β_1 sits inside three different exponentials. But try $\beta_1 = \log 2$, so $e^{\beta_1} = 2$: $S = 34 - 2 - 2 \cdot 4 - 3 \cdot 8 = 0$. That is the answer; Newton finds it without the lucky guess.

Regressors Inside: β_1 by Hand, the Iterations ★

Slope over curvature, starting from $\beta_1 = 0.5$. At each guess, first the four fitted means $\theta_i = e^{\beta_1 X_i}$, then S and $-\ell''$ from the previous frame:

guess β_1	$\theta_i = e^{\beta_1 X_i}$	slope S	$-\ell''$	step	next
0.500	1, 1.65, 2.72, 4.48	13.47	52.9	0.255	0.755
0.755	1, 2.13, 4.53, 9.63	-6.06	106.9	-0.057	0.698
0.698	1, 2.01, 4.04, 8.12	-0.45	91.3	-0.005	0.693

First row, spelled out: $S = 34 - 1.65 - 2(2.72) - 3(4.48) = 13.47$ and $-\ell'' = 1.65 + 4(2.72) + 9(4.48) = 52.9$, so the step is $13.47/52.9 = 0.255$.

$\hat{\beta}_1 = 0.693 = \log 2$: the fitted means 1, 2, 4, 8 match the data exactly. $e^{\hat{\beta}_1} = 2$: each unit of X doubles the expected count. The final curvature, 90.0, gives $se = 1/\sqrt{90} = 0.105$. With an intercept too, the same steps run on two equations at once; that is what `glm(y ~ X, family = poisson)` does.

Practical Matters in R

- `optim()` **minimizes**. MLE is a maximization problem, so you pass it the *negative* log-likelihood.
- Supply `par`, the starting values. Poor starting values are the most common cause of failed convergence.
- Always check the convergence code (`$convergence == 0`), a returned number is not the same as a converged number.
- Ask for `hessian = TRUE`. The *Hessian* is the matrix of second derivatives of the function you optimized, computed numerically at the solution; with one parameter it is the single number $-\ell''(\hat{\theta})$, the observed information. Invert it for the variance; square-root for the standard error.
- Try several starting values. If they land in different places, your likelihood is multimodal or your model is not identified.

All of this is Thursday's lab.

Identification

Definition

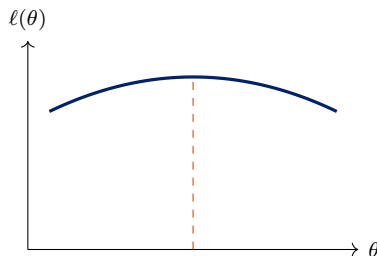
θ is **identified** if distinct parameter values imply distinct distributions of the data:
 $\theta^* \neq \theta \Rightarrow f(\cdot | \theta^*) \neq f(\cdot | \theta)$.

Non-identification means the likelihood is *flat* along some direction: the data cannot distinguish among a set of parameter values.

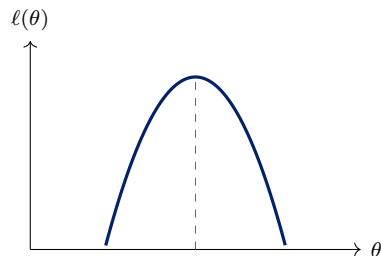
- **Classic example:** $Y_i \sim \mathcal{N}(\theta_1 - \theta_2, 1)$. Only the *difference* is identified. Optimizers started at different points will return different $(\hat{\theta}_1, \hat{\theta}_2)$ lying along a ridge, all with the same log-likelihood.
- This is the likelihood analogue of **perfect collinearity** in OLS, and it has the same symptom: a singular Hessian, so no standard errors.
- Diagnosis: multiple start values, and inspecting the Hessian's rank.

Information

Curvature Is Information ★



flat: little information



peaked: much information

Both log-likelihoods peak at the same $\hat{\theta}$. But on the left, many nearby values of θ fit almost as well; we are uncertain. On the right, moving away from $\hat{\theta}$ costs a lot of fit; we are confident.

The second derivative measures exactly this: the more negative, the more information.

Fisher Information ★

Definition (one parameter, at the true value)

$$\mathcal{I}(\theta_0) \equiv -E[\ell''(\theta_0)] = E[S(\theta_0)^2] = \text{Var}(S(\theta_0)).$$

Mind the direction: the true θ_0 is fixed and the data vary, Week 2's sampling-distribution view, which is how any estimator is judged (estimation ran the other way). E and Var average over $y \sim f(y \mid \theta_0)$. Minus sign because $\ell'' < 0$ at a peak.

- The second and third forms are the **information equality**, $-E[\ell''] = \text{Var}(S)$: proved in the orange frames, true if the model is right. It lets one number, the curvature at the peak, give the standard error.
- *Additive:* ℓ is a sum of n like terms, so $-E[\ell'']$ is too: $\mathcal{I}_n = n\mathcal{I}_1$, the familiar $1/n$.
- **From data:** plug $\hat{\theta}$ into your sample's own $-\ell''(\theta)$. This is the *observed information*, what software reports.

The Variance of the MLE ★

$$\hat{V}(\hat{\theta}) = \left[-\frac{\partial^2 \ell}{\partial \theta^2} \Big|_{\hat{\theta}} \right]^{-1} = \frac{1}{\text{observed information}}$$

Read it as a sentence: *the variance is one over minus the second derivative of the log-likelihood at its peak.* **Why**, in one Newton step from the truth: the score is zero at the peak, so

$$0 = S(\hat{\theta}) \approx S(\theta_0) + \ell''(\theta_0)(\hat{\theta} - \theta_0) \implies \hat{\theta} - \theta_0 \approx \frac{S(\theta_0)}{-\ell''(\theta_0)}.$$

The score varies across samples with variance $\mathcal{I}$ (the information equality); dividing by the curvature $\mathcal{I}$ leaves $\text{Var}(\hat{\theta}) \approx \mathcal{I}/\mathcal{I}^2 = 1/\mathcal{I}$. The orange frames fill in the steps. This is why `optim(..., hessian = TRUE)` is not optional.

Several parameters ★: the observed information is a matrix (the Hessian, sign flipped); invert it, and the standard errors are the square roots of its diagonal.

Information Equality: Why Care ★

The Variance frame's Newton step: $\hat{\theta} - \theta_0 \approx S(\theta_0)/(-\ell''(\theta_0))$. The denominator is a sum over observations, so by the law of large numbers (as in Week 2) replace it by its expectation, a fixed number $1/c = -E[\ell''(\theta_0)]$. Then $\text{Var}(cS) = c^2 \text{Var}(S)$:

$$\text{Var}(\hat{\theta}) \approx \frac{\text{Var}(S(\theta_0))}{(-E[\ell''(\theta_0)])^2}.$$

These are two different quantities. The information equality, $\text{Var}(S(\theta_0)) = -E[\ell''(\theta_0)] = \mathcal{I}(\theta_0)$, says they are the same number, so the ratio collapses to $\mathcal{I}/\mathcal{I}^2 = 1/\mathcal{I}$.

- One number, the curvature at the peak, is the whole standard error. Without the equality you would need both pieces separately: Week 2's sandwich.
- The $\approx$ is a first-order expansion, exact in the $\sqrt{n}$ limit (Properties section).

Information Equality: Goal and Plan ★

What we want to show. Setting, in three lines: one observation $y \sim f(y \mid \theta)$; its own score $s(y; \theta) \equiv \partial \log f(y \mid \theta) / \partial \theta = (\partial f / \partial \theta) / f$, evaluated at that same true θ ; only y is random, and E averages over it. The full score is $S = \sum_i s(y_i; \theta)$. Then:

$$-E \left[\frac{\partial^2 \log f}{\partial \theta^2} \right] = E[s^2] = \text{Var}(s).$$

The only tool. The density integrates to one at *every* θ , so all its θ -derivatives vanish (we may swap $d/d\theta$ and $\int$, one of the “regularity conditions”):

$$\int f(y \mid \theta) dy = 1 \quad \text{for all } \theta \quad \implies \quad \frac{d}{d\theta} \int f(y \mid \theta) dy = 0.$$

Plan.

1. Differentiate once $\Rightarrow E[s] = 0$.
2. Differentiate twice $\Rightarrow E[(\partial^2 f / \partial \theta^2) / f] = 0$.
3. Quotient rule on $s \Rightarrow \partial^2 \log f / \partial \theta^2 = (\partial^2 f / \partial \theta^2) / f - s^2$.
4. Take expectations of step 3; steps 1–2 leave only s^2 .

Information Equality: Steps 1–2 ★

Step 1: differentiate once.

$$\begin{aligned}
 0 &= \frac{d}{d\theta} \int f(y \mid \theta) dy && \text{derivative of the constant 1} \\
 &= \int \frac{\partial f(y \mid \theta)}{\partial \theta} dy && \text{move the derivative inside the integral} \\
 &= \int s(y; \theta) f(y \mid \theta) dy && s = (\partial f / \partial \theta) / f, \text{ so } \partial f / \partial \theta = s f; s \text{ depends on } y \text{ too} \\
 &= E[s(y; \theta)] && \int g(y) f(y \mid \theta) dy = E[g(y)] \text{ for } y \sim f(\cdot \mid \theta), \text{ this same } \theta
 \end{aligned}$$

Step 2: differentiate twice. Line two above is 0 at every θ ; differentiate it again:

$$\begin{aligned}
 0 &= \int \frac{\partial^2 f(y \mid \theta)}{\partial \theta^2} dy && \text{derivative of a zero is zero} \\
 &= \int \frac{\partial^2 f / \partial \theta^2}{f} f dy = E \left[\frac{\partial^2 f / \partial \theta^2}{f} \right] && \text{multiply and divide by } f: \text{ an expectation}
 \end{aligned}$$

Information Equality: Step 3 ★

Step 3: differentiate the score. $s = (\partial f / \partial \theta) / f$ is a ratio, so use the quotient rule $(u/v)' = (u'v - uv')/v^2$ with $u = \partial f / \partial \theta$ and $v = f$:

$$\frac{\partial^2 \log f}{\partial \theta^2} = \frac{\partial s}{\partial \theta}$$

s is the first derivative of $\log f$

$$= \frac{\frac{\partial^2 f}{\partial \theta^2} f - \frac{\partial f}{\partial \theta} \frac{\partial f}{\partial \theta}}{f^2}$$

$u' = \partial^2 f / \partial \theta^2$, $v' = \partial f / \partial \theta$

$$= \frac{\partial^2 f / \partial \theta^2}{f} - \left(\frac{\partial f / \partial \theta}{f} \right)^2$$

split the fraction; cancel one f in each piece

$$= \frac{\partial^2 f / \partial \theta^2}{f} - s^2$$

the bracket is s by definition

An identity in y and θ , no expectation yet. The first term is the quantity step 2 showed has mean zero; the second is the squared score.

Information Equality: Step 4 ★

Step 4: take expectations of step 3.

$$\begin{aligned}
 E\left[\frac{\partial^2 \log f}{\partial \theta^2}\right] &= E\left[\frac{\partial^2 f / \partial \theta^2}{f}\right] - E[s^2] && \text{expectation of a difference} \\
 &= 0 - E[s^2] && \text{step 2 kills the first term} \\
 &= -\text{Var}(s) && \text{step 1: } E[s] = 0, \text{ so } \text{Var}(s) = E[s^2] - 0^2
 \end{aligned}$$

Multiply by -1 : $-E[\partial^2 \log f / \partial \theta^2] = \text{Var}(s)$. Done, for one observation.

- With n independent observations, $\ell'' = \sum_i \partial^2 \log f(y_i | \theta) / \partial \theta^2$ and $S = \sum_i s(y_i; \theta)$: expectations add, and variances of independent terms add, so $-E[\ell''] = \text{Var}(S)$ as well.
- Every expectation was taken under the *same* f that integrates to one. If the data come from some other g , steps 1 and 2 fail, the two sides part ways, and the Hessian standard error is wrong.

Worked: Bernoulli Information

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1 - \theta},$$
$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} - \frac{n - \sum_i y_i}{(1 - \theta)^2}.$$

Taking expectations, using $E[\sum_i y_i] = n\theta$:

$$\mathcal{I}(\theta) = \frac{n\theta}{\theta^2} + \frac{n(1 - \theta)}{(1 - \theta)^2} = \frac{n}{\theta} + \frac{n}{1 - \theta} = \frac{n}{\theta(1 - \theta)}.$$

Worked: Bernoulli Information

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Taking expectations, using $E[\sum_i y_i] = n\theta$:

$$\mathcal{I}(\theta) = \frac{n\theta}{\theta^2} + \frac{n(1 - \theta)}{(1 - \theta)^2} = \frac{n}{\theta} + \frac{n}{1 - \theta} = \frac{n}{\theta(1 - \theta)}.$$

$$\text{Var}(\hat{\theta}) = \mathcal{I}(\theta)^{-1} = \frac{\theta(1 - \theta)}{n}. \quad \checkmark$$

Exactly the variance of a sample proportion you already know, now *derived* from the shape of the likelihood.

Worked: Poisson Information

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} \implies \mathcal{I}(\theta) = \frac{E[\sum_i y_i]}{\theta^2} = \frac{n\theta}{\theta^2} = \frac{n}{\theta},$$

so

$$\text{Var}(\hat{\theta}) = \frac{\theta}{n}, \quad \hat{V}(\hat{\theta}) = \frac{\bar{y}}{n}.$$

Check directly: $\hat{\theta} = \bar{y}$ and $\text{Var}(\bar{Y}) = \text{Var}(Y)/n = \theta/n$, because the Poisson mean equals its variance. ✓ Two routes, same answer; when a direct calculation exists, use it as a check.

Properties

The Four Properties

Under regularity conditions, as $n \rightarrow \infty$:

1. **Consistency** ★: $\hat{\theta} \xrightarrow{p} \theta_0$.
2. **Asymptotic normality** ★: $\hat{\theta} \overset{a}{\sim} \mathcal{N}(\theta_0, \mathcal{I}(\theta_0)^{-1})$.
3. **Asymptotic efficiency** ★: if the model is right, no other well-behaved estimator has a smaller asymptotic variance (the **Cramér–Rao bound**). Stronger than Gauss–Markov, which ranked only *linear* unbiased estimators.
4. **Invariance** ★: if $\hat{\theta}$ is the MLE of θ , then $c(\hat{\theta})$ is the MLE of $c(\theta)$ for any function $c(\cdot)$. Wonderfully convenient: the MLE of the odds is the odds of the MLE.

The first three are asymptotic, none of them says anything about $n = 40$. Invariance, by contrast, holds exactly in any sample.

Why 1–2 hold: the Newton step from the truth (Variance frame), $\hat{\theta} - \theta_0 \approx S(\theta_0)/(-\ell''(\theta_0))$, has a sum in the numerator (CLT: normal) and a sum in the denominator (LLN: converges to $\mathcal{I}$), structurally Week 2's OLS argument ★.

Maximum Likelihood Is Not Unbiased ★

- We already saw it: $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_i \hat{e}_i^2$ has $E[\hat{\sigma}^2] = \frac{n-2}{n} \sigma^2$ in the bivariate model; s^2 divides by $n - 2$ precisely to undo this.

The general claim. Maximum likelihood makes no promise about unbiasedness in a finite sample. Some MLEs happen to be unbiased (the Poisson $\hat{\theta} = \bar{y}$; $\hat{\beta}$ in the normal linear model, which is OLS); others are biased ($\hat{\sigma}^2$ above). What the method guarantees is that $\hat{\theta}$ closes in on θ_0 as n grows (consistency); whatever bias there is at $n = 40$ is a finite-sample matter that vanishes with it.

The trade we are making

Give up exact unbiasedness in finite samples; get consistency, asymptotic efficiency, and a completely general recipe that works for any probability model you can write down.

Testing

Goal and Plan

What we want. A test of a hypothesis about θ that uses nothing but the log-likelihood curve, and a clear reading of which test R ran when it printed a p -value.

The plan.

1. A null hypothesis is one value, θ_0 . The question: is θ_0 far from the peak $\hat{\theta}$?
2. The obvious ruler, $z = (\hat{\theta} - \theta_0)/\text{se}$, works at large n and breaks at small n near a boundary.
3. The curve offers two ways to measure “far”: the horizontal gap (Wald) and the vertical drop in fit (likelihood ratio). On a parabola (i.e., with large n) they are one number.
4. Off a parabola they differ. Compute both on Simulation 2’s samples: which one still works at small n ?
5. Which test your software ran, and what to report in practice.

A Null Hypothesis Is a Smaller Model ★

Week 2 tested $\beta_j = 0$. Look closely at what that null is.

- $\beta_j = 0$ is the model *without* X_j : the same regression with one parameter pinned at a fixed value.
- In general, $\theta = \theta_0$ is the model with θ pinned at θ_0 , the **restricted** model. The model where θ is free is the **unrestricted** one; its best value is $\hat{\theta}$.

So these are the same question:

- Is $\hat{\beta}_j$ statistically significant?
- Does the model with X_j fit better than the model without it?

Is the null value θ_0 far from the peak $\hat{\theta}$?

One question. The log-likelihood curve will give three ways to measure “far.” Start with the one you already know.

The Obvious Ruler: z ★

The last section gave $\hat{\theta}$ and $\text{se}(\hat{\theta}) = 1/\sqrt{\text{observed information}}$. Week 2's test divided an estimate by its standard error, so:

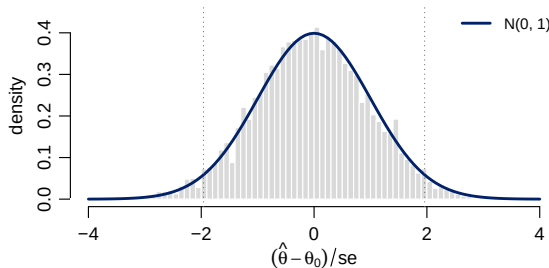
$$z = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}, \quad \text{reject if } |z| > 1.96.$$

- This is the **Wald test**. It is what `summary(glm)` prints next to every coefficient.

Does it work? Check it the way Week 2 checked everything: simulate under the null and count rejections. Two settings, one comfortable, one not.

Simulation 1: z at Large n ★

Poisson counts, true $\theta_0 = 2$, $n = 200$. Draw 20,000 samples; in each, compute $\hat{\theta} = \bar{y}$, $\text{se} = \sqrt{\hat{\theta}/n}$ (Worked: Poisson Information), and $z = (\hat{\theta} - \theta_0)/\text{se}$.

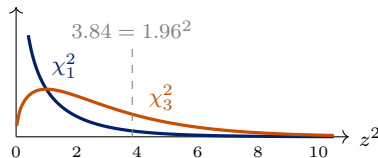


- The histogram sits on the $\mathcal{N}(0, 1)$ curve.
- Reject when $|z| > 1.96$: 5.1% of samples, as advertised.

Verdict: at large n the ruler works. The normal limit from the Properties section is doing exactly what it promised.

Before the uncomfortable setting, one change of units that the other ruler will need.

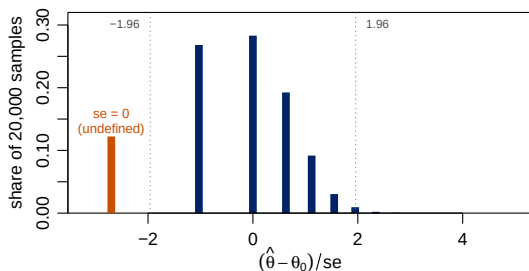
Square It: z^2 and the χ^2



- $|z| > 1.96$ is the same event as $z^2 > 1.96^2 = 3.84$. Same test, written on the square.
- Under the null z is standard normal, so z^2 has a fixed distribution too. Its name is χ_1^2 , “chi-squared with one degree of freedom.” Its 95% point is 3.84 by construction.
- Squares add. A null that pins q parameters at once gives q squared z ’s summed: χ_q^2 , mean q , cut-offs 3.84, 5.99, 7.81 for $q = 1, 2, 3$.

Simulation 2: Small n , Near a Boundary ★

Bernoulli trials, $\theta_0 = 0.1$, $n = 20$, $se = \sqrt{\hat{\theta}(1 - \hat{\theta})/n}$ (Worked: Bernoulli). Draw 20,000 samples; each is s successes out of 20, so z takes only a few values.



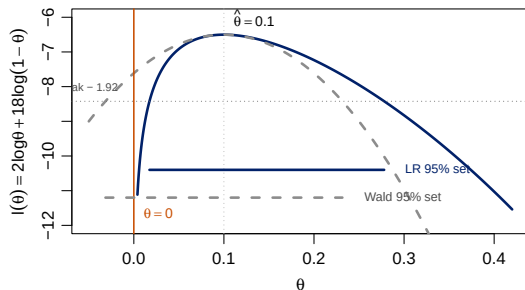
- $s = 0$ happens in 12% of samples. Then $\hat{\theta} = 0$, $se = 0$: z does not exist.
- The lowest finite z is -1.03 ($s = 1$: $\hat{\theta} = 0.05$, $se = 0.049$). The test can *never* reject from below.
- It rejects 0.2% of the time, not 5%.

Verdict: the ruler breaks. Why? z uses two numbers, the peak and its width, and nothing else. What curve is it reading? Take one sample and draw it.

Simulation 2: the Curve z Is Reading

One sample, $s = 2$ of $n = 20$: $\ell(\theta) = 2\log\theta + 18\log(1 - \theta)$, peak $\hat{\theta} = 0.1$, $\text{se} = 0.067$ (Worked frames).

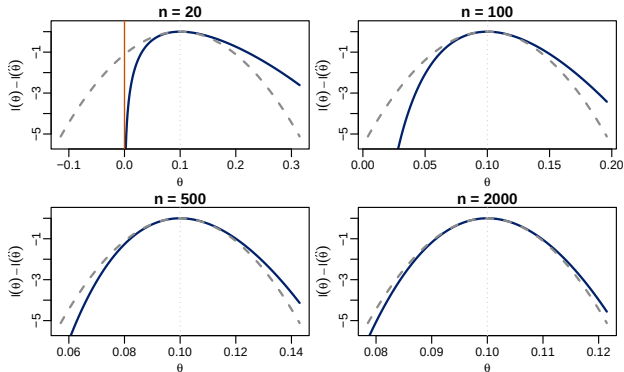
What z ever sees is a peak and a curvature of a parabola $\ell(\hat{\theta}) - 1/2 \cdot \mathcal{I}(\theta - 0.1)^2$.



- Solid: the real ℓ . Dashed: the parabola z reads.
- **Left of the peak** the real curve dives to $-\infty$ at $\theta = 0$ (two successes are impossible there); the parabola sails past into negative probabilities.

Wald's 95% set, $0.1 \pm 1.96 \times 0.067 = (-0.03, 0.23)$, contains impossible values. Is this a small- n disease? Watch the same curve as n grows.

Large n Makes a Parabola: Watch It Happen



Bernoulli with $\hat{\theta} = 0.1$: the real ℓ (solid) and the parabola with the same peak and curvature (dashed), each panel $\hat{\theta} \pm 3$ SEs.

From $n = 20$ to $n = 2000$ the solid curve straightens into the dashed parabola, exactly over the window a test looks at: a few SEs either side of the peak. Why? Next frame.

Why Large n Makes a Parabola ★

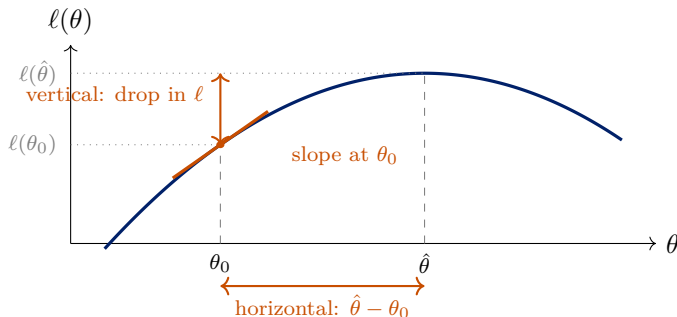
Expand ℓ around its peak one term past the parabola, and look c standard errors away, at $\theta = \hat{\theta} + c \text{ se}$:

$$\begin{aligned} \ell(\hat{\theta} + c \text{ se}) - \ell(\hat{\theta}) &= -\frac{1}{2} \mathcal{I} (c \text{ se})^2 + \frac{1}{6} \ell'''(\hat{\theta}) (c \text{ se})^3 + \cdots && \text{Taylor; slope 0 at the peak} \\ &= -\frac{c^2}{2} + \frac{c^3}{6} \cdot \frac{\ell'''(\hat{\theta})}{\mathcal{I}^{3/2}} + \cdots && \text{se} = 1/\sqrt{\mathcal{I}} \end{aligned}$$

- The first term is the parabola: at c SEs from the peak it drops by $c^2/2$, whatever n is.
- The second term is the bend away from the parabola. ℓ is a sum of n terms, so ℓ''' grows like n , while $\mathcal{I}^{3/2}$ grows like $n^{3/2}$: the ratio shrinks like $1/\sqrt{n}$.

So within a few SEs of the peak, which is the only place a test looks, the drop is $c^2/2$ plus an error that vanishes as n grows: the last frame's panels. Next: what exactly a test reads off this curve.

Three Rulers ★



- **Horizontal:** the gap $\hat{\theta} - \theta_0$, in standard-error units. *Wald: the z-test*
- **Vertical:** the fit lost by moving from $\hat{\theta}$ down to θ_0 . *Likelihood ratio*
- **Slope at θ_0 :** at a peak it would be zero. A third ruler; we use it in Week 4. *Score*

Two Rulers: the Formulas ★

Each is a squared z , so each is compared with χ_1^2 : reject above 3.84.

- **Wald: the gap over its standard error, squared.** $W = z^2 = \mathcal{I}(\hat{\theta} - \theta_0)^2$, where $\mathcal{I} = -\ell''(\hat{\theta})$ is the observed information and $\text{se} = 1/\sqrt{\mathcal{I}}$ (Variance frame).
- **Likelihood ratio: the drop in fit, doubled.** $LR = 2[\ell(\hat{\theta}) - \ell(\theta_0)]$. Never negative, because the peak is the highest point. Why doubled: next frame.

With q restrictions: $LR = 2[\ell(\hat{\theta}_{\text{unrestricted}}) - \ell(\hat{\theta}_{\text{restricted}})] \xrightarrow{d} \chi_q^2$, and likewise W . The one-parameter case has $\hat{\theta}_{\text{restricted}} = \theta_0$.

On a Parabola, the Two Rulers Are One Number

Suppose ℓ is the parabola with peak $\hat{\theta}$ and curvature $-\mathcal{I}$:

$$\ell(\theta) = \ell(\hat{\theta}) - \frac{1}{2} \mathcal{I} (\theta - \hat{\theta})^2 \quad \text{the parabola: peak at } \hat{\theta}, \text{ curvature } -\mathcal{I}$$

$$\ell(\hat{\theta}) - \ell(\theta_0) = \frac{1}{2} \mathcal{I} (\hat{\theta} - \theta_0)^2 \quad \text{plug in } \theta = \theta_0: \text{ the drop}$$

$$LR = 2 \times \text{drop} = \mathcal{I} (\hat{\theta} - \theta_0)^2 = W \quad \text{doubling undoes the } \frac{1}{2}$$

- So $W = LR$: one distance, two names, one cut-off.
- Read the LR line backwards: $LR > 3.84$ is $\text{drop} > 1.92$, Simulation 2's dotted line. "1.96 SEs from the peak" and "1.92 below the peak" are the same set of θ 's.

So the rulers agree exactly when ℓ is a parabola. Simulation 2's curve was not one: compute both on it.

Simulation 2, Continued: Both Rulers

Same setting, both rulers. Exact rejection rates at 3.84, true $\theta_0 = 0.1$:

n	W undefined	W rejects	LR rejects
20	12.2%	0.2%	13.3%
50	0.5%	11.6%	5.8%
100	0	6.8%	4.4%
500	0	5.7%	5.2%

- $n = 20$: neither ruler is at 5%. The $s = 0$ samples show why: W does not exist, and $LR = 2[0 - 20 \log 0.9] = 4.21$ rejects a sample that occurs 12% of the time under the null. Two expected successes: no parabola; use an exact binomial test.
- $n = 50$: W 's failure flips. One or two successes give a tiny se, so a huge $|z|$: W rejects 11.6%. LR is near 5%.
- $n \geq 100$: both settle toward 5%, W the slower: LR reads the curve, W extrapolates it. Each row down is the parabola arriving, as the panels showed.

Which Test Your Software Ran ★

At large n the two rulers agree. In a small sample or near a boundary (Simulation 2) they do not, and it matters which one you are reading.

In your output

- `summary(glm)` z -statistics and p -values: **Wald**.
- `anova(m0, m1, test = "Chisq")`, the deviance, “does adding this block improve the fit”: **likelihood ratio**.

When Wald and LR disagree, believe LR: it measured the curve, z extrapolated it. The next frame says which one to reach for.

In Practice: What to Report ★

The working rules

- **One coefficient, large n , an ordinary variable:** the Wald z and p that `summary()` prints are fine. That is what published tables report, and at large n it equals LR.
- **A block:** dummies for a categorical variable, an interaction, a set of controls, “does adding these help.” Fit both models and use LR: `anova(m0, m1, test = "Chisq")`. No single z answers that question (Lab 3, region).
- **A rare outcome, a small group, a huge standard error, a coefficient near a boundary:** do not trust z . Refit without the term and report LR.

The habit to build: when a z looks strange, ask what curve it read.